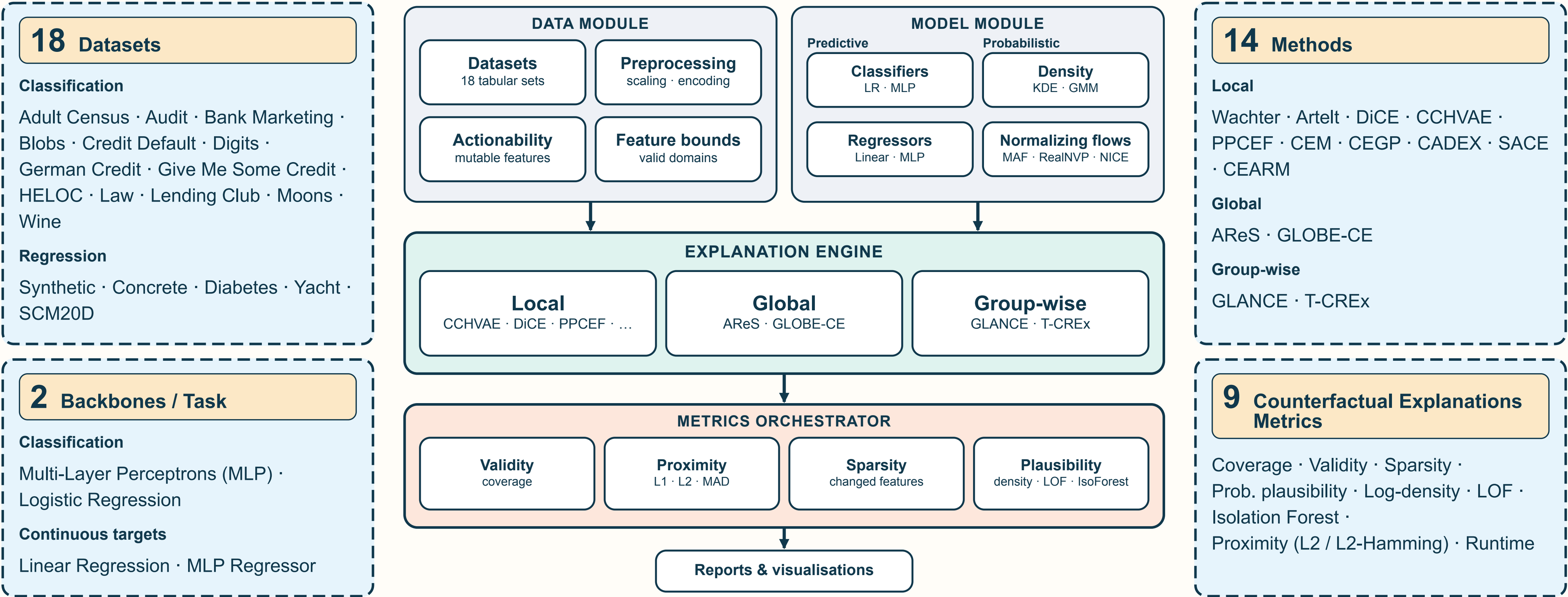
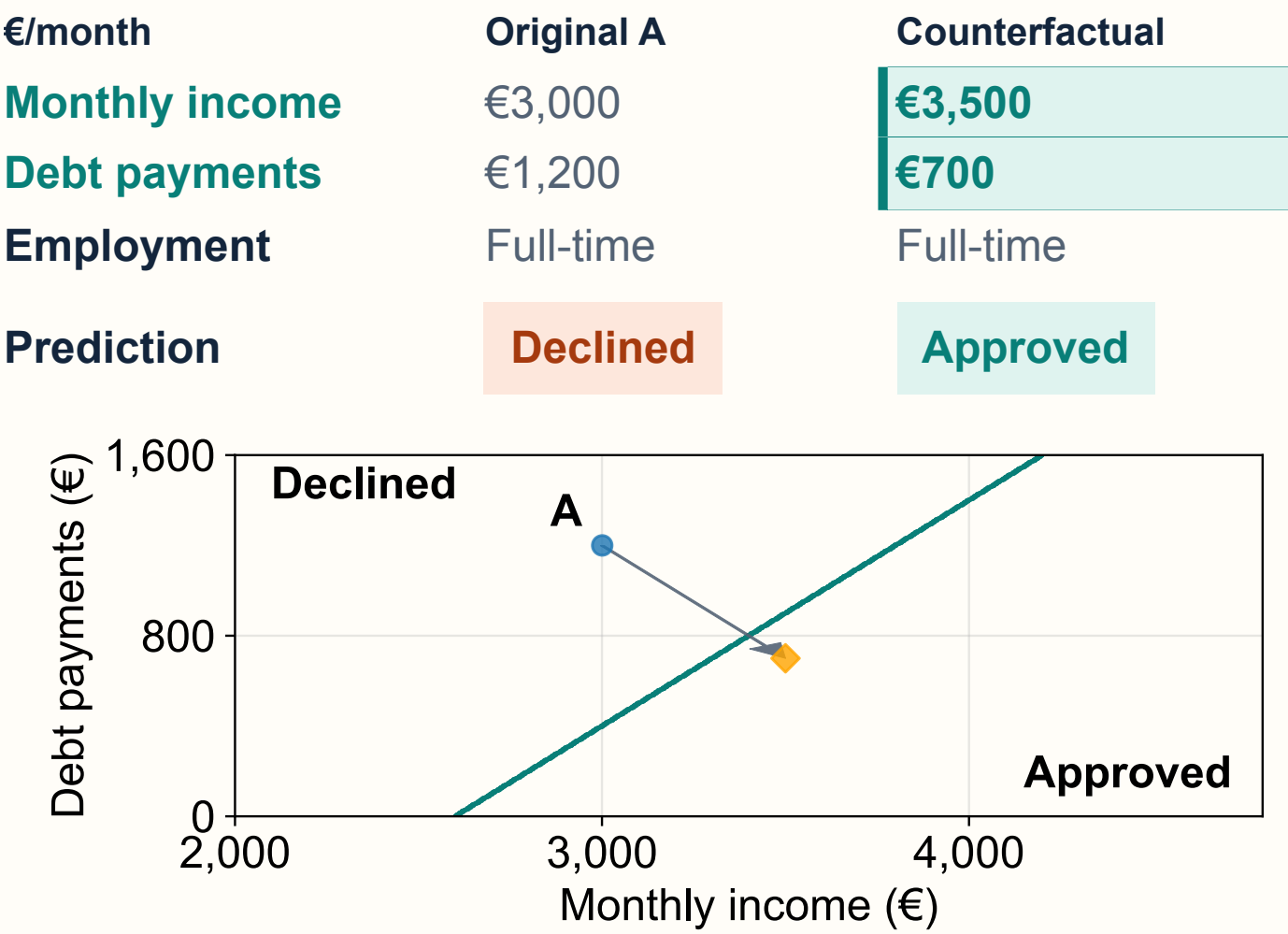




What is a counterfactual explanation? A small input change that flips a model's prediction.

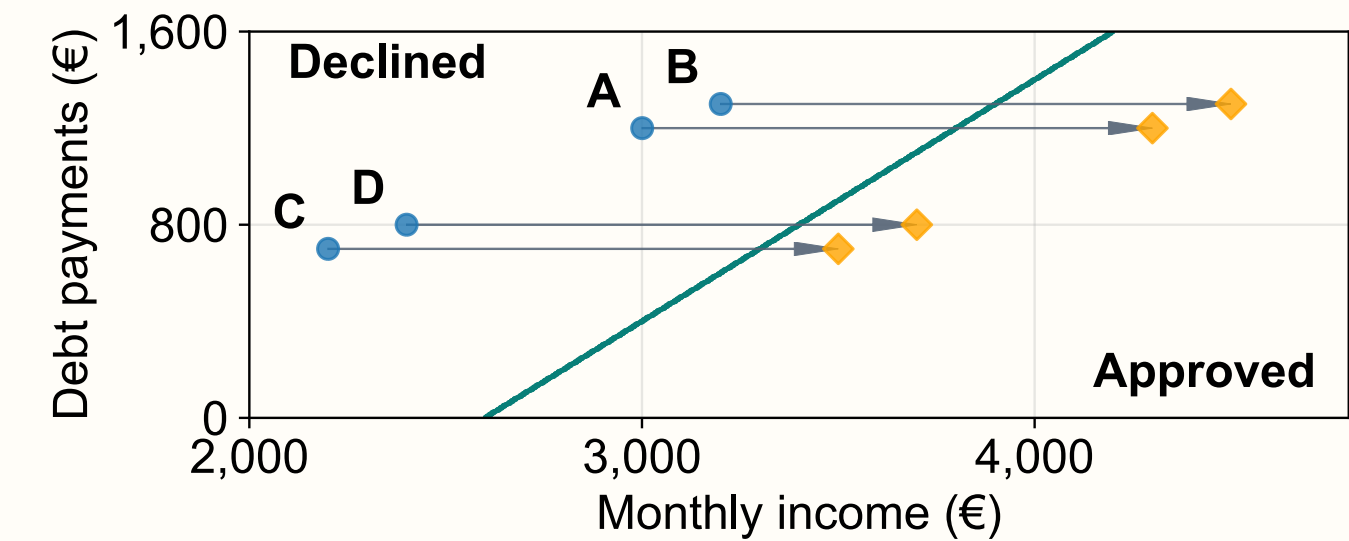
LOAN APPLICATION EXAMPLE

Local One applicant



Global One shared change

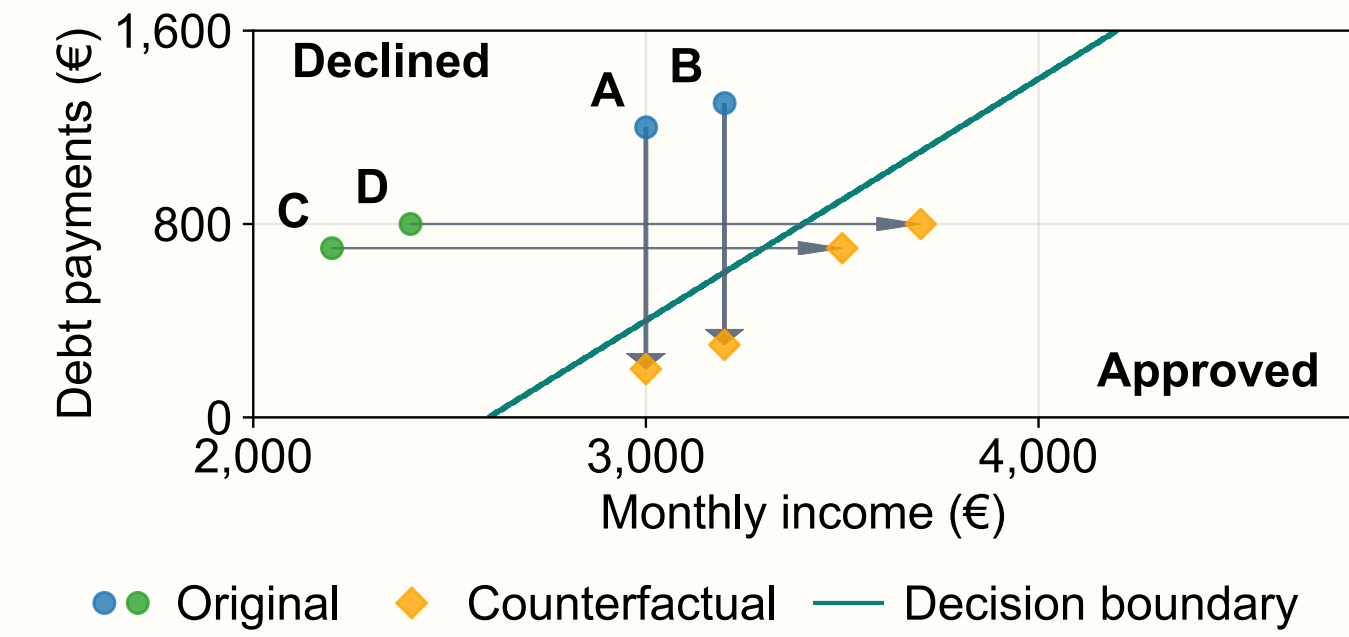
A-D Income +€1,300; debt unchanged.



Group-wise One change per group

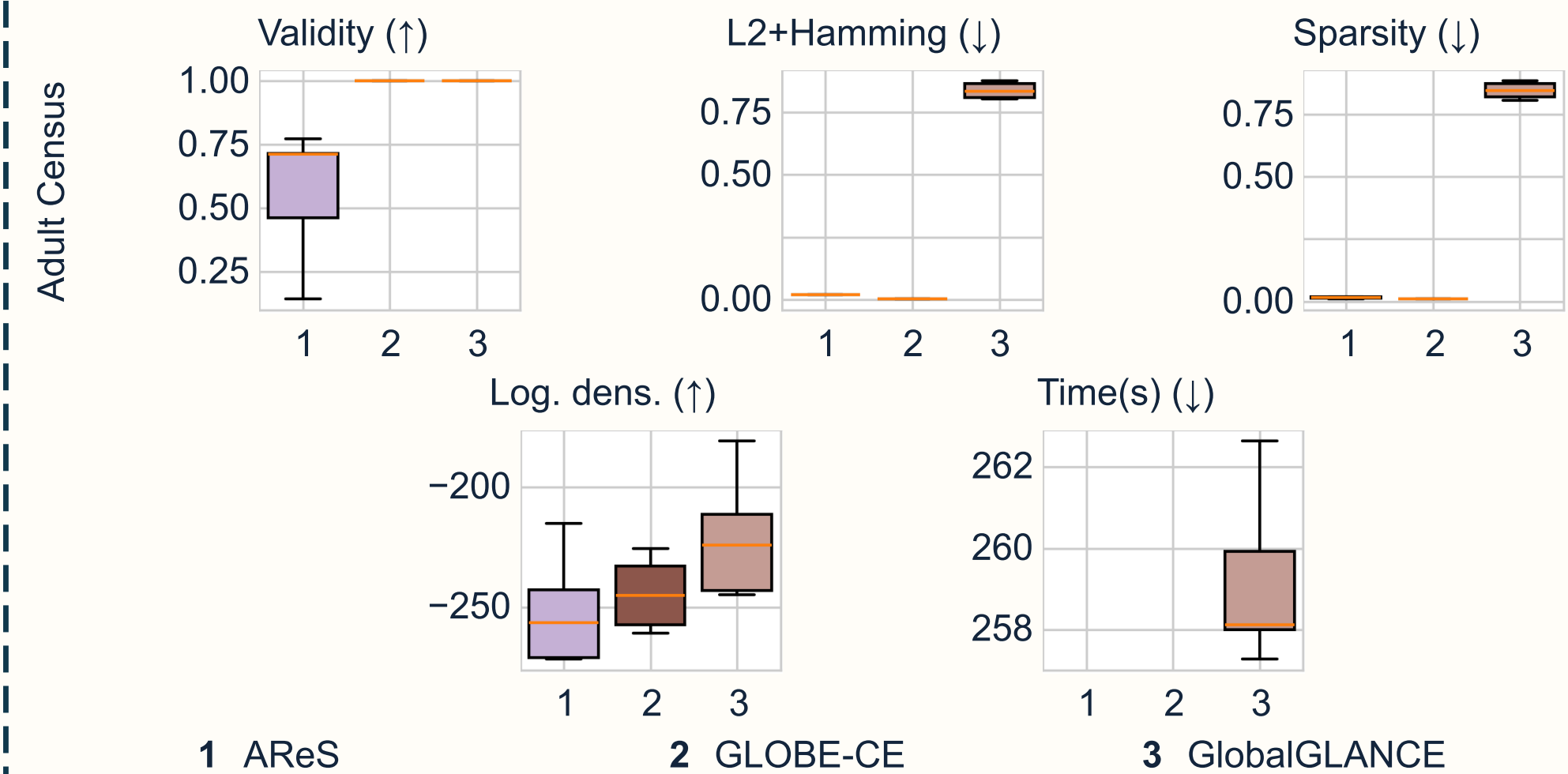
A-B Debt -€1,000; income unchanged.

C-D Income +€1,300; debt unchanged.

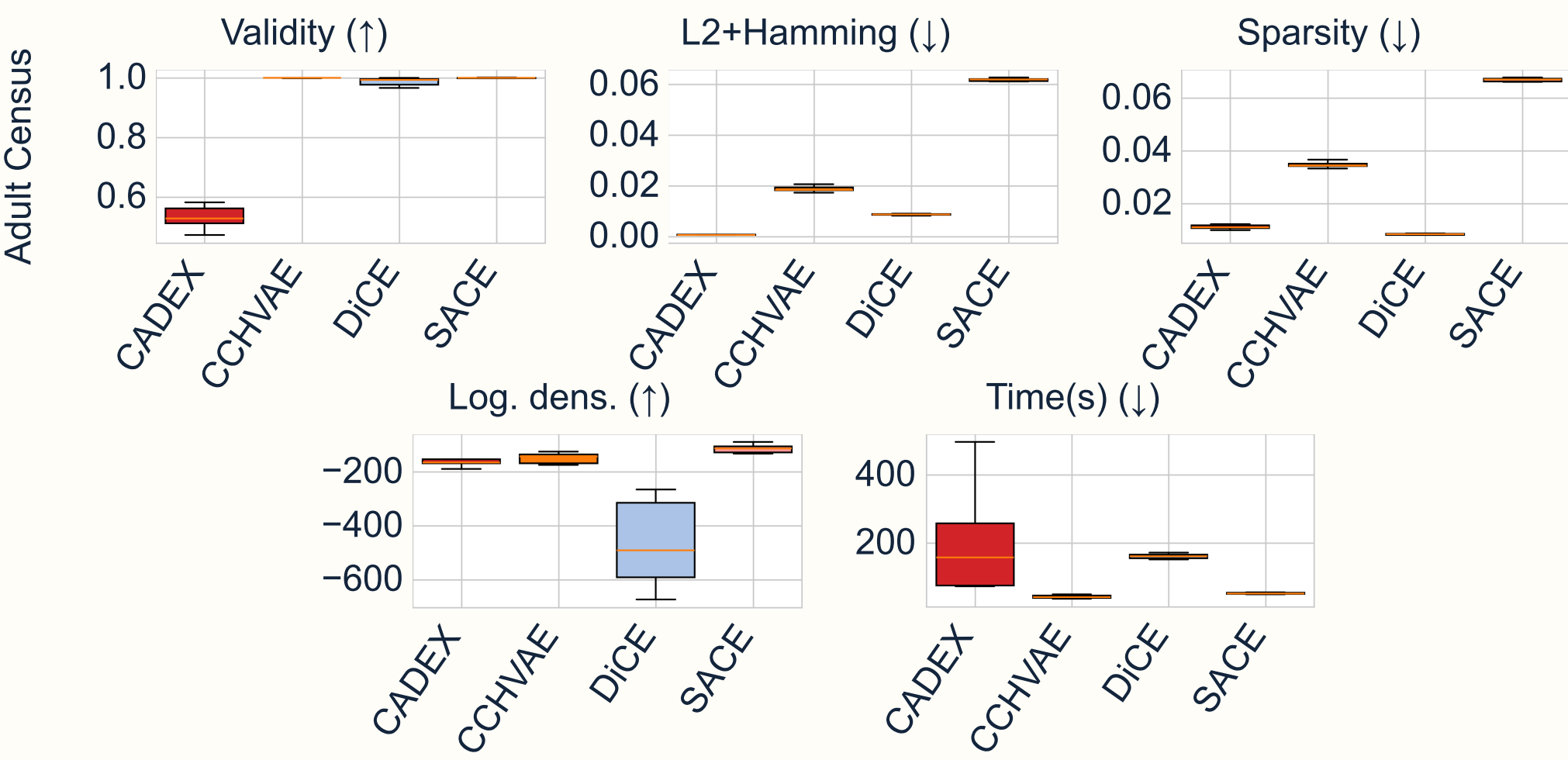


Results

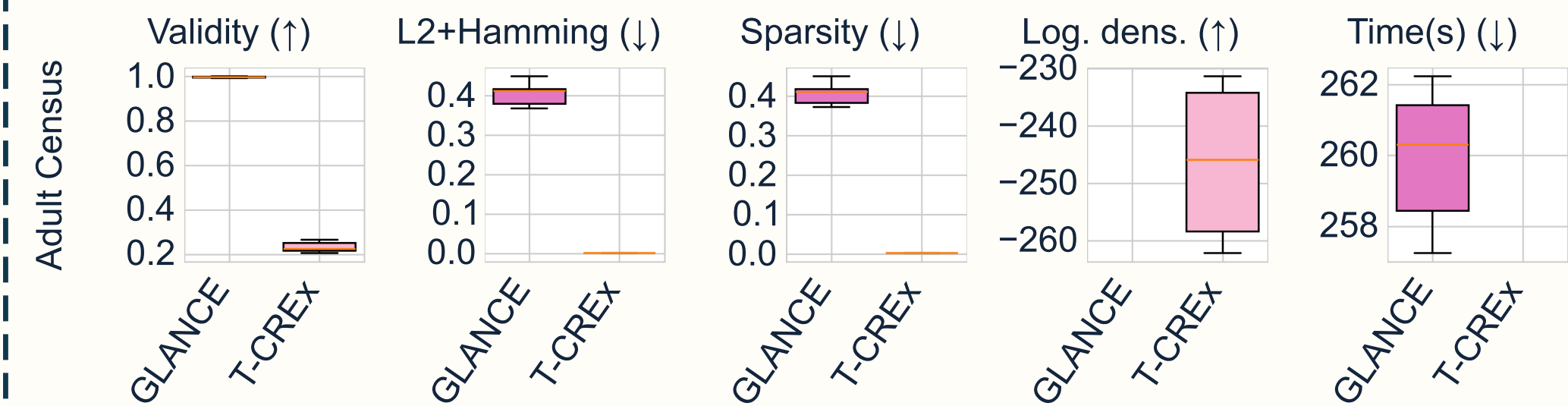
Global: validity and failures



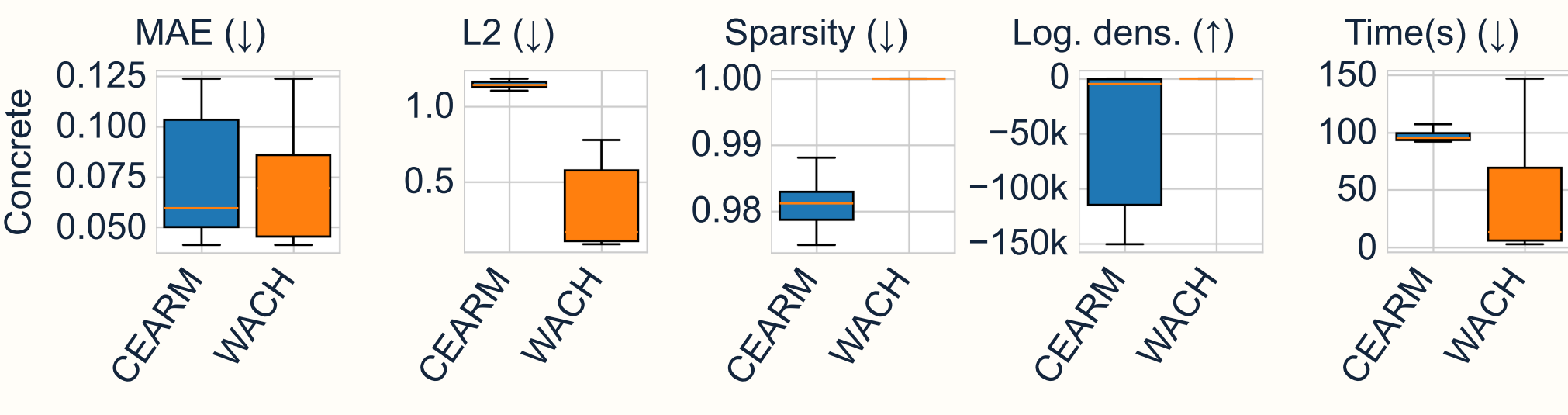
Local: no single winner



Group-wise: reach versus change



Regression: accuracy versus change



Contributions

01 Controlled protocol

Reproducible evaluation

02 Cross-paradigm benchmark

Local · Global · Group-wise

03 Extensible library

Open source

